First Assignment

Agent-Based Modeling Task

Collective Intelligence - Autumn 2026

Semester 2026/27/1

Task Description

Create a **custom NetLogo model** of a social, biological, or physical phenomenon you want to study. Inspiration from the NetLogo Models Library is allowed, but your submission must be **novel in logic, agents, and experiments**.

1. Objective

- Develop a unique NetLogo model to simulate a clearly stated phenomenon and **research question/hypothesis**.
- You may browse library models for ideas, but you must **not** base your solution on small tweaks of a library model (see "Banned bases").

Example research questions:

- "How does the ratio of explorers to exploiters affect a colony's food collection efficiency when resources are patchy vs. uniform?"
- "At what population density does a rumor transition from dying out to spreading through an entire network with varying trust thresholds?"

2. Model Requirements

- The environment is a **2D grid** with horizontal and vertical wrapping.
- **Map size**: choose between 20×20 and 128×128.
- **Interface** includes:
  - **At least 5** adjustable hyperparameters (Sliders/Switches/Choosers/Inputs).
  - Buttons: setup, go-once, go.
  - **At least 3** reporters (Monitors/Plots), including one time series and one distribution (e.g., histogram).
- **Model structure** must include:
  - **Two or more breeds** that interact (e.g., competition, exchange, predation, cooperation).
  - At least **3 agent attributes** and **3 global variables**.
  - **One dynamic environment process** (e.g., regenerating resource, diffusing signal, changing topology).
  - **At least one** non-trivial feature:
    - state-dependent or nonlinear rule (thresholds, saturation, feedback), or

    * simple local learning/adaptation (e.g., reinforcement of a movement preference), or
    * communication/signaling between agents, or
    * endogenous creation/removal of patches/resources.
  - Clean code with **helper procedures** and comments. Avoid monolithic go.
- **Documentation:**
  - Fill the Info Tab describing: (1) what the model does and why, (2) how agents behave and interact, (3) what each parameter controls.

3. Experiment Requirements

- Use **BehaviorSpace** to run a **two-factor** experiment:
  - Sweep two hyperparameters across ranges that plausibly include a phase transition or tipping point.
  - **Repetitions:** 20 per configuration (random seed varied).
  - **Reporters:** choose **two** outcome measures (e.g., mean cluster size, extinction time).
- **Export** results to CSV and compute (in any tool) **mean** and **95% confidence intervals** for each cell in the grid of settings.
- **Plots:** include (i) a heatmap or contour for the 2D sweep, and (ii) a time series illustrating a representative run in each regime. Clearly indicate any estimated tipping point.

Mini Report (max 2 pages, PDF)

- **Page 1 (Model summary):** What does the model do? What agents exist and how do they behave? What are the key parameters?
- **Page 2 (Experiments & Results):** Two-factor design, parameter ranges, figures (heatmap with mean $\pm$ 95% CI and time series), identification of regime change, one-sentence takeaway and limitations.

PowerPoint Presentation

Presenting is optional, but **not presenting caps your maximum score at 20 out of 40 points**. If you present (recommended 5–6 minutes, 5–6 slides):

Content

- **Introduction:** phenomenon, question, hypothesis.
- **Model**: breeds, rules, key interface elements.
- **Demonstration:** GIF/short video of the model.
- **Experiments:** BehaviorSpace design, heatmap/CI plots, interpretation of tipping point.

Assignment Submission and General Rules

Submission Files (one ZIP)

- .nlogo model (self-contained).
- .xml BehaviorSpace configuration.
- **Mini report (PDF, max 2 pages)**.
- **Presentation** in PDF AND PPTX (if presenting).

Banned Bases

No small tweaks allowed: Fire, Wolf Sheep Predation, Segregation, Ants, Traffic Basic, Virus, and close lookalikes. If inspired by them, you must **reformulate** the problem with different entities, rules, and outcomes. Superficial reskins will receive a 0.

Academic Integrity

- Code similarity checks will be used. Submissions with substantial overlap with library models, past solutions, or between students will receive **0**.
- Not submitting anything results in **0**. Submitting a non-empty attempt that fails requirements may earn at least **1**.

Grading Rubric

Total: 40 points

All requirements must be met for full credit.

Originality & Modeling (12 pts)

- Two interacting breeds ..... **4 pts**
- Dynamic environment ..... **3 pts**
- At least one non-trivial mechanism ..... **3 pts**
- Coherent ODD summary ..... **2 pts**

Implementation Quality (8 pts)

- Clean modular code and comments ..... **3 pts**
- Interface completeness (buttons, 5+ controls) ..... **3 pts**
- Required monitors/plots incl. histogram ..... **2 pts**

Experimentation (12 pts)

- Correct two-factor sweep ..... **4 pts**
- 20 reps with seed variation ..... **3 pts**

- Mean ± 95% CI reported ..... **3 pts**
- Clear heatmap & regime identification ..... **2 pts**

Analysis & Interpretation (5 pts)

- Justified parameter ranges ..... **1 pt**
- Clear narrative of results & tipping point ..... **3 pts**
- Limitations/future work ..... **1 pt**

Presentation (3 pts)

- If presented: clarity, timing, and insight ..... **3 pts**
- Without presentation, maximum score is capped at 20/40.

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